

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 1: Information Representation

Subtopic 1.1: Data Representation

Concept 1.1.3: Binary Coded Decimal (BCD)

Total Questions: 11

Mark Schemes begin on Page 5

Question 1 | Source: 9618_w25_qp_11 (Q1.a.ii)

(ii) Convert the denary number into Binary Coded Decimal (BCD).

108

..... [1]

Question 2 | Source: 9618_w25_qp_12 (Q7.c.i)

(c) ASCII is another character set. The ASCII value for the character 'h' has the denary value 104.

(i) Write the Binary Coded Decimal (BCD) value for the ASCII character 'h'.

..... [1]

Question 3 | Source: 9618_s25_qp_12 (Q2.c)

(c) Give **one** application where Binary Coded Decimal (BCD) is used and justify its use.

Application

Justification

..... [2]

Question 4 | Source: 9618_w24_qp_11 (Q1.b.iii)

(iii) Convert the Binary Coded Decimal (BCD) into denary.

010101110011

Answer [1]

Question 5 | Source: 9618_w23_qp_12 (Q3.b.ii)

(ii) Convert the Binary Coded Decimal (BCD) into denary.

100001100101

Answer [1]

Question 6 | Source: 9618_w23_qp_12 (Q3.c)

(c) Identify **one** practical application of BCD **and** justify why BCD is used in this application.

Application

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Justification

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[2]

Question 7 | Source: 9618_s23_qp_11 (Q3.d.ii)

3 A computer has an Operating System (OS).

(d) The computer stores data in binary form.

(ii) Convert the denary number 964 into Binary Coded Decimal (BCD).

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..... [1]

Question 8 | Source: 9618_w22_qp_11 (Q1.a.ii)

(ii) Convert the Binary Coded Decimal (BCD) into denary.

00100111

Answer [1]

Question 9 | Source: 9618_w22_qp_12 (Q2.a.iii)

(iii) Convert the unsigned binary integer into Binary Coded Decimal (BCD). Show your working.

10010101

Working

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Answer [2]

Question 10 | Source: 9618_w22_qp_13 (Q9.b)

(b) State **two** benefits of using Binary Coded Decimal (BCD) to represent values.

Benefit 1

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Benefit 2

.....

[2]

Question 11 | Source: 9618_w21_qp_12 (Q4.d)

4 A register stores the following binary number:

1	1	0	0	1	1	0	1
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(d) State why the value in the register cannot be interpreted as a Binary Coded Decimal (BCD).

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..... [1]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q1.a.ii)

1(a)(ii)	0001 0000 1000	1
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Mark Scheme for Question 2 | Source: 9618_w25_ms_12 (Q7.c.i)

7(c)(i)	1 mark for the correct answer 0001 0000 0100	1
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Mark Scheme for Question 3 | Source: 9618_s25_ms_12 (Q2.c)

2(c)	1 mark for application and 1 mark for corresponding justification e.g. • an application that performs financial / banking calculations • because financial transactions use only two decimal places and must be accurate, no accumulating / rounding errors and it is difficult to represent decimal values exactly in normal binary. Or • electronic displays • because visual displays only need to show individual digits • because conversion between denary and BCD is straightforward Or • The storage of the date and time in the BIOS of a PC • because conversion between denary and BCD is more straightforward Or • Barcode systems • because conversion between denary and BCD can be accurately completed	2
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Mark Scheme for Question 4 | Source: 9618_w24_ms_11 (Q1.b.iii)

1(b)(iii)	1 mark for: 573	1
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Mark Scheme for Question 5 | Source: 9618_w23_ms_12 (Q3.b.ii)

3(b)(ii)	865	1
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Mark Scheme for Question 6 | Source: 9618_w23_ms_12 (Q3.c)

3(c)	1 mark for a correct application 1 mark for a corresponding justification - An application that performs financial / banking calculations ... because it is difficult to represent decimal values exactly in normal binary and financial transactions use only two decimal places and must be accurate, no accumulating errors Or - Electronic displays, e.g. calculators, digital clocks ... because visual displays only need to show individual digits ... because conversion between denary and BCD is easier Or - The storage of the date and time in the BIOS of a PC ... because conversion with denary is easier	2
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Mark Scheme for Question 7 | Source: 9618_s23_ms_11 (Q3.d.ii)

3(d)(ii)	1001 0110 0100	1
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Mark Scheme for Question 8 | Source: 9618_w22_ms_11 (Q1.a.ii)

1(a)(ii)	27	1
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Mark Scheme for Question 9 | Source: 9618_w22_ms_12 (Q2.a.iii)

2(a)(iii)	1 mark for each bullet point: 149 decimal 0001 0100 1001	2
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Mark Scheme for Question 10 | Source: 9618_w22_ms_13 (Q9.b)

9(b)	1 mark for each correct benefit (max 2) Examples: - straightforward to convert to / from BCD and denary ... so it is less complex to encode and decode for programmers - easier for digital equipment use BCD to display output information - can represent monetary values exactly	2
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Mark Scheme for Question 11 | Source: 9618_w21_ms_12 (Q4.d)

4(d)	1 mark for: The denary value in each group of 4 bits is greater than 9 // the denary value in each nibble is greater than 9	1
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